

CHEMISTRY | UNIT 03 | LESSON 04

Metallic Bonding and Material Properties

Independent Application Answer Sheet

Name: _____ Date: _____

MLA standards: MLA.CHEM.BON.01 and MLA.CHEM.LAB.03

Directions: The observed property data are provided below. Study each row and complete only the bonding-explanation column. Then answer all three analysis questions. Submit the corrected sheet with Notebook Evidence.

Part A - Property evidence table

Sample	Conductivity	Malleability	Luster	Student bonding explanation
Copper	High	High; bends	Shiny	
Aluminum	High	High; rolls into sheets	Shiny	
Sulfur	Very low	None; brittle	Dull	

Part B - Analysis

1. Which two samples show metallic properties? Cite at least two observations. 2. Explain conductivity using mobile delocalized valence electrons. 3. Explain malleability using sliding metal-ion layers and continued attraction to the electron cloud.

Question	Student response
1	
2	
3	